Review of Finite-Horizon MDPs.

- optimal value func $V^{\pi^*}(s) = \max_{a \in A} R(s,a) + \gamma \mathbb{E}_{s' \sim P_{sa}} [V^{\pi^*}(s')]$
- optimal policy $\pi^*(s) = \operatorname{argmax}_{a \in A} R(s,a) + \gamma \mathbb{E}_{s' \sim P_{sa}} [V^{\pi^*}(s')]$

Modification

- $(S, A, P_{sa}^{(t)}, T, R^{(t)})$
- payoff $= R(s_0, a_0) + R(s_1, a_1) + \cdots + R(s_T, a_T) \neq \sum_{t=0}^{\infty} R(s_t, a_t) \gamma^t$ → 무한히 발산하지 않도록 하는 장치

$$\left| \sum_{t}^{\infty} R(s_t) \gamma^t \right| \leq \bar{R} \sum_{t=0}^{\infty} \gamma^t$$

- non-stationary policy $\pi^{(t)} : S \to A$ (time, State, Action) instant $+1$ vs long-term $+\infty$
- transition distribution & reward $s_{t+1} \sim P_{sa}^{(t)}$ (time) $R(s_t, a_t)^{(t)}$ (time)

- value function $V_t(s) = \mathbb{E}[R^{(t)}(s_t, a_t) + \cdots + R^{(T)}(s_T, a_T) \mid s_t = s, \pi]$ (time)

(i) $t = T$: $V_T^*(s) = \max_{a \in A} R^{(T)}(s,a)$ (action)

(ii) $0 \leq t < T$: $V_t^*(s) = \max_{a \in A} \left[R^{(t)}(s,a) + \mathbb{E}_{s' \sim P_{sa}^{(t)}} [V_{t+1}^*(s')] \right]$ (reward, action)

$$\| V_{t+1} - V^* \|_\infty = \| B(V_t) - V^* \|_\infty \leq \gamma \| V_t - V^* \|_\infty \leq \gamma^t \| V_1 - V^* \|_\infty$$

- $V_{t+1}(s) = \max_{a \in A} \left[R(s,a) + \gamma \sum_{s'} P_{sa}(s') V_t(s') \right]$
- $V^*(s) = \max_{a \in A} \left[R(s,a) + \gamma \sum_{s'} P_{sa}(s') V^*(s') \right]$

Therefore Bellman operator is γ-contracting operator

- $|V_{t+1}(s) - V^*(s)| \leq \max_{a \in A} \gamma \sum_{s'} P_{sa}(s') |V_t(s') - V^*(s')|$

Linear Quadratic Regulator

LQR ① system dynamics ⇒ linear

② cost function ⇒ quadratic

Note

- zero-mean noise 은 optimal policy와 무관.
- $U_t = W_t = I_d$ then $R_t = -\|s_t\|^2 - \|a_t\|^2$ (regularization)

- linear $s_{t+1} = A_t s_t + B_t a_t + w_t$ $\quad s_t \in \mathbb{R}^d \quad a_t \in \mathbb{R}^d \quad w_t \sim N(0, \Sigma_t)$
- quad $R^{(t)}(s_t, a_t) = -s_t^T U_t s_t - a_t^T W_t a_t < 0$ (U_t, W_t: positive definite.)

Step 1. estimate A, B, Σ

(i) collect (s_t, a_t, s_{t+1})

(ii) linear regression $\arg\min_{A,B} \sum_i \sum_t \| s_{t+1}^{(i)} - (A s_t^{(i)} + B a_t^{(i)}) \|^2$

(iii) $w_t^{(i)} = s_{t+1}^{(i)} - A s_t^{(i)} - B a_t^{(i)}$

$\Sigma = \frac{1}{N} \sum_i w_t^{(i)} (w_t^{(i)})^T$

Step 2. Derive optimal policy (DP)

- $s_{t+1} = A s_t + B a_t + w_t$
- $R^{(t)}(s_t, a_t) = -s_t^T U_t s_t - a_t^T W_t a_t$
- $V_T^*(s_T) = \max_{a_T} R_T(s_T, a_T)$

$= \max_{a_T} -s_T^T U_T s_T - a_T^T W_T a_T$ ($a_T = 0$)

Lemma 1. V_{t+1}^* quad function in s_t then V_t^* is also quadratic

- if $V_{t+1}^*(s_{t+1}) = s_{t+1}^T \Phi_{t+1} s_{t+1} + \Psi_{t+1}$
- then $V_t^*(s_t) = s_t^T \Phi_t s_t + \Psi_t$

Lemma 2. optimal policy is just linear func of state. i.e $a_t^* = L_t s_t$

Lemma 3.

- Φ_t, L_t depends on neither Ψ nor noise Σ_t
- Ψ_t, V_t^* depends on Σ_t

- (linear) $s_{t+1} = A_t s_t + B_t a_t + w_t$

- $\mathbb{E}[s_{t+1}^T \Phi_{t+1} s_{t+1}] = (A_t s_t + B_t a_t)^T \Phi_{t+1} (A_t s_t + B_t a_t)$

$$+ \underbrace{\mathbb{E}[w_t^T \Phi_{t+1} w_t]}_{\because \mathrm{tr}(\Sigma_t \Phi_{t+1}) \text{ because } w_t \sim N(0, \Sigma_t)}$$

- $V_t^*(s_t) = \max_{a_t} \left[R^{(t)}(s_t, a_t) + \mathbb{E}_{s_{t+1} \sim P^{(t)}_{s_t a_t}} \left[V_{t+1}^*(s_{t+1}) \right] \right]$

$R^{(t)}(s_t, a_t)$ → $-s_t^T U_t s_t - a_t^T W_t a_t$

$\mathbb{E}_{s_{t+1}} [s_{t+1}^T \Phi_{t+1} s_{t+1} + \Psi_{t+1}]$ → $\sim N(A_t s_t + B_t a_t, \Sigma_t)$

- $J(a_t) = -s_t^T U_t s_t - a_t^T W_t a_t + (A_t s_t + B_t a_t)^T \Phi_{t+1} (A_t s_t + B_t a_t) + \mathrm{tr}(\Sigma_t \Phi_{t+1}) + \Psi_{t+1}$

- $\frac{\partial J}{\partial a_t} = -2 W_t a_t + 2 B_t^T \Phi_{t+1} (A_t s_t + B_t a_t) = 0$

- $a_t^* = (W_t - B_t^T \Phi_{t+1} B_t)^{-1} B_t^T \Phi_{t+1} A_t s_t$

- $\Phi_t = A_t^T (\Phi_{t+1} - \Phi_{t+1} B_t (B_t^T \Phi_{t+1} B_t - W_t)^{-1} B_t^T \Phi_{t+1}) A_t - U_t$

- $\Psi_t = \mathrm{tr}(\Sigma_t \Phi_{t+1}) + \Psi_{t+1}$

---

linearize the system and use LQR

- State $\bar{s}_t$ 에서 테일러의 식같을 생성.
- action $\bar{a}_t$ 근처의 동작을 추론 함.

- $s_t \approx F(\bar{s}_t, \bar{a}_t) + \nabla_s F(\bar{s}_t, \bar{a}_t)(s_t - \bar{s}_t) + \nabla_a F(\bar{s}_t, \bar{a}_t)(a_t - \bar{a}_t)$

$F(\bar{s}_t, \bar{a}_t) \because k$, $\nabla_s F(\bar{s}_t, \bar{a}_t) \because A$, $\nabla_a F(\bar{s}_t, \bar{a}_t) \because B$

Differential Dynamic Programming (trajectory of rockets)

(i) come up with trajectory that approximate the one we want to follow

(ii) linearize around trajectory point s_t^*

(iii) use LQR to find optimal policy π_t

(iv) with π_t produce new trajectory

Step 1. sample trajectory $s_0^*, a_0^* \rightarrow s_1^*, a_1^* \rightarrow s_2^*, a_2^* \cdots$

Step 2. Linearize dynamics

- $s_{t+1} \approx F(s_t^*, a_t^*) + \nabla_s F(s_t^*, a_t^*) \cdot (s_t - s_t^*) + \nabla_a F(s_t^*, a_t^*) \cdot (a_t - a_t^*)$

$= K + A s_t + B a_t$

$= A s_t + B a_t$ (add one-dim to s_t)

- $R(s_t, a_t) \approx R(s_t^*, a_t^*) + \frac{1}{2}(s_t - s_t^*)^T H_{ss} (s_t - s_t^*) + \nabla_s R(s_t^*, a_t^*)(s_t - s_t^*) + (s_t - s_t^*)^T H_{sa} (a_t - a_t^*) + \nabla_a R(s_t^*, a_t^*)(a_t - a_t^*) + \frac{1}{2}(a_t - a_t^*)^T H_{aa} (a_t - a_t^*)$

$= -s_t^T U_t s_t - a_t^T W_t a_t$

- $\begin{bmatrix} 1 & x \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} = a + 2bx + cx^2$

Step 3. find optimal policy with LQR

- $s_{t+1} = A_t s_t + B_t a_t$
- $R^{(t)}(s_t, a_t) = -s_t^T U_t s_t - a_t^T W_t a_t$

Step 4. produce new trajectory

- $s_0^* \; \pi(s_0^*) \rightarrow s_1^*, \pi(s_1^*) \rightarrow \cdots \rightarrow s_T^*$
- use real $s_{t+1}^* = F(s_t^*, \pi_t(s_t^*))$

Partially Observable MDPs.

- $O_t \mid S_t \sim O(o \mid s)$ (observation / state) e.g. object in camera { 20% car, 80% human }
- Finite-horizon POMDPs : (S, O, A, P_{sa}, T, R)

Gaussian sensor noise / compression

$\Rightarrow$ Linear Quadratic Gaussian (LQG)

$y_t = C \cdot s_t + v_t$

$s_{t+1} = A \cdot s_t + B \cdot a_t + w_t$

$s_t \mid y_1, \dots, y_t \sim N(s_{t|t}, \Sigma_{t|t})$

---

suppose no action dependency

$$\begin{cases} s_{t+1} = A s_t + w_t \sim N(0, \Sigma_s) \\ y_t = C \cdot s_t + v_t \sim N(0, \Sigma_y) \end{cases}$$

- $[s_1, \dots, s_t, y_1, \dots, y_t] \sim N(\mu, \Sigma)$
- $s_t \mid y_1, \dots, y_t \sim N(s_{t|t}, \Sigma_{t|t})$ using marginal formula.

Note. Marginal Formula

- $\begin{bmatrix} x_a \\ x_b \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix} \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix} \right)$
- $x_a \mid x_b \sim N(\mu_{a|b}, \Sigma_{a|b})$
- $\mu_{a|b} = \mu_a + \Sigma_{ab} \Sigma_{bb}^{-1} (x_b - \mu_b)$
- $\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab} \Sigma_{bb}^{-1} \Sigma_{ba}$.

Kalman Filter $s_t \mid y_1, \dots, y_t \sim N(s_{t|t}, \Sigma_{t|t})$

(i) compute $s_{t+1} \mid y_1, \dots, y_t \sim N(s_{t+1|t}, \Sigma_{t+1|t})$

- $s_{t+1|t} = A s_{t|t}$
- $\Sigma_{t+1|t} = A \Sigma_{t|t} A^T + \Sigma_s$

Note.

$$\begin{bmatrix} s_{t+1} \\ y_{t+1} \end{bmatrix} \sim N\left(\begin{bmatrix} s_{t+1|t} \\ C s_{t+1|t} \end{bmatrix}, \begin{bmatrix} \Sigma_{t+1|t} & \Sigma_{t+1|t} C^T \\ C\Sigma_{t+1|t} & C \Sigma_{t+1|t} C^T + \Sigma_y \end{bmatrix} \right)$$

(ii) compute $s_{t+1} \mid y_1, \dots, y_{t+1} \sim N(s_{t+1|t+1}, \Sigma_{t+1|t+1})$

- $s_{t+1|t+1} = s_{t+1|t} + K_t (y_{t+1} - C \cdot s_{t+1|t})$
- $\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K_t C \Sigma_{t+1|t}$
- Kalman Gain

$$K_t = \Sigma_{t+1|t} C^T (C \Sigma_{t+1|t} C^T + \Sigma_y)^{-1}$$

t 이전의 observation 필요.

distribution $s_t | (y_1, \dots, y_t)$

$\Rightarrow$ mean $s_{t|t}$ to be

best appx for s_t

$\Rightarrow$ action $a_t := L_t s_{t|t}$ with LQR

Note.

- $s_{t|t}$ is noisy appx of s_t
- LQR is independent of noise.

LQG Summary

(i) forward pass $\Rightarrow$ $\mu_t, \Sigma_{t|t}, s_{t|t}$

(ii) backward pass $\Rightarrow$ Ψ_t, Φ_t, L_t

(iii) optimal policy $a^*_t = L_t s_{t|t}$.

Policy Gradient

- do not need value / Q functions
- trajectory $\tau = (s_0, a_0, s_1, a_1, \cdots, s_T)$ with $T < \infty$
- randomized policy $\pi_\theta(a|s)$
- sample $\{P_{sa}\}$ $\Rightarrow$ do not need
  energy $R(s,a)$ analytic form

- $s_0 \sim$ distribution μ $\Rightarrow$ maximize $\eta(\theta) = \mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t R(s_t, a_t)\right]$
- $a_t \sim \pi_\theta(\cdot|s_t)$ $= \mathbb{E}_{s_0 \sim \mu}\left[V^{\pi_\theta}(s_0)\right]$
- $s_t \sim P_{s_t a_{t-1}}$

- distribution $\tau \sim P_\theta(\tau)$
- $f(\tau) = \sum_{t=0}^{T-1} \gamma^t R(s_t, a_t)$
- $\eta(\theta) = \mathbb{E}_{\tau \sim P_\theta}[f(\tau)]$ $\Rightarrow$ P_θ는 θ에 의존한다. (VAE랑 비슷)

$$\nabla_\theta \mathbb{E}_{\tau \sim P_\theta}[f(\tau)] = \nabla_\theta \int P_\theta(\tau) f(\tau) d\tau$$

$$= \int (\nabla_\theta P_\theta(\tau)) f(\tau) d\tau$$

$$= \int P_\theta(\tau) (\nabla_\theta \log P_\theta(\tau)) f(\tau) d\tau$$

$$= \mathbb{E}_{\tau \sim P_\theta}[(\nabla_\theta \log P_\theta(\tau)) f(\tau)] \approx \frac{1}{N} \sum_{i=1}^{N} (\nabla_\theta \log P_\theta(\tau)) f(\tau^{(i)})$$

- $\log P_\theta = \log \mu(s_0) + \log \pi_\theta(a_0|s_0) + \cdots + \log P_{s_{T-1} a_{T-1}}(s_T)$
- $\nabla_\theta \log P_\theta = \nabla_\theta \log \pi_\theta(a_0|s_0) + \cdots + \nabla_\theta \log \pi_\theta(a_{T-1}|s_{T-1})$

$$\Rightarrow \nabla_\theta \eta(\theta) = \nabla_\theta \mathbb{E}_{\tau \sim P_\theta}[f(\tau)] = \mathbb{E}_{\tau \sim P_\theta}\left[\left(\underbrace{\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)}_{\text{direction}}\right)\left(\underbrace{\sum_{t=0}^{T-1} \gamma^t R(s_t, a_t)}_{\text{payoff}}\right)\right]$$

- $\mathop{\mathbb{E}}_{a\sim\pi(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s)\right] = \sum_a \pi_\theta(a|s)\cdot \nabla_\theta \log \pi_\theta(a|s)$

$= \sum_a \nabla_\theta \pi_\theta(a|s)$

$= \nabla_\theta \sum_a \pi_\theta(a|s)$

$= \nabla_\theta 1$

$= 0$

- $\mathop{\mathbb{E}}_{\tau\sim P_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\right] = \sum_{t=0}^{T-1} \mathop{\mathbb{E}}_{\tau\sim P_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\right]$ ($= 0$)

intuition : $f(\tau) = 1$ $\quad = 0$

- $\mathop{\mathbb{E}}_{\tau\sim P_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot B(s_t)\right] = \mathop{\mathbb{E}}_{\tau\sim P_\theta}\left[B(s_t)\cdot \mathbb{E}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\,|\,s_0, a_0, s_1, \ldots, s_t\right]\right]$

이미 고정됨. $\quad = 0$

- $\mathop{\mathbb{E}}_{\tau\sim P_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{j=0}^{T-1} \nu^j R(s_j,a_j)\right)\right]$

$= \sum_{0\le j<t} \nu^j R_j + \sum_{t\le j} \nu^j R_j$

과거 보상 $\quad$ 현재 이후 보상

Note.

B는 a_t가 뽑히기 전의

정보 $H_t = (t, s_0, a_0, \ldots, a_{t-1}, s_t)$를

인과로 조합 가능

with $\mathop{\mathbb{E}}_{\tau\sim P_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{0\le j<t} \nu^j R_j(s_j,a_j)\right)\right] = 0$

이미 고정됨.

- $\nabla_\theta \eta(\theta) = \sum_{t=0}^{T-1} \mathop{E}_{\tau\sim P_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot\left(\sum_{j=t}^{T-1} \nu^j R(s_j,a_j) - r^t B(s_t)\right)\right]$

$= \sum_{t=0}^{T-1} \mathop{E}_{\tau\sim P_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot \nu^t\left(\sum_{j=t}^{T-1} \nu^{j-t} R(s_j,a_j) - B(s_t)\right)\right]$

- baseline $B(\cdot)$는 이항에 variance ↓
- $B(s_t) := \mathbb{E}\left[\sum_{j=t}^{T-1} \nu^{j-t} R(s_j,a_j)\,|\,s_j\right]$

$= V^{\pi_\theta}(s_t)$

Note.

precise value of $V^{\pi_\theta}(\cdot)$

does not influence mean

of the estimator

Vanilla Policy Gradient with Baseline

(i) collect set of trajectories by executing current policy

(ii) $R_{\geq t} = \sum_{j=t}^{T-1} \gamma^{j-t} R(s_j, a_j)$

(iii) baseline B

minimizes $\sum_k \sum_t \left(R_{\geq t} - B(s_t) \right)^2$

(iv) update policy param θ

with GD $\sum_k \sum_t \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \left(R_{\geq t} - B(s_t) \right)$